

BRAINSTORMING & IDEA GENERATION

EDUGENIE – AI-POWERED LEARNING ASSISTANT

Team Members:

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1. Project Context

EduGenie is conceived as an AI-powered learning assistant that brings several common learning-support activities into one web application.

2. Learning Problems Identified

Students may need help at different stages of learning:

- understanding a question;
- simplifying a difficult concept;
- creating practice questions;
- revising long educational material;
- deciding what to study first and what to study next.

3. Initial Idea

The initial project idea is to build one learning assistant with separate modules for Question Answering, Concept Explanation, Quiz Generation, Text Summarization, and Learning Path Generation.

4. Why an AI-Based Approach

The documented project uses generative AI so that the application can produce responses dynamically rather than relying only on fixed educational content.

5. Proposed User Experience:

A learner opens the EduGenie interface, chooses a learning function, provides the required input, submits the request, and receives the generated result through the web interface.

6. AI Model Strategy

The documented implementation uses Gemini 3.5 Flash for cloud-based text generation functions and LaMini-Flan-T5-783M locally for concept explanation.

7. Modular Idea

Each major learning function is kept in a separate Python module. This allows the project to keep question answering, explanation, quiz generation, summarization, and learning-path logic separated.

8. Expected Value

EduGenie is intended to provide a single, straightforward interface for multiple academic-assistance tasks.

9. Expected Outcome

The final application should demonstrate multiple AI-assisted learning workflows connected through a FastAPI backend.